NOTE: The document identifier and heading has been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

INCH-POUND

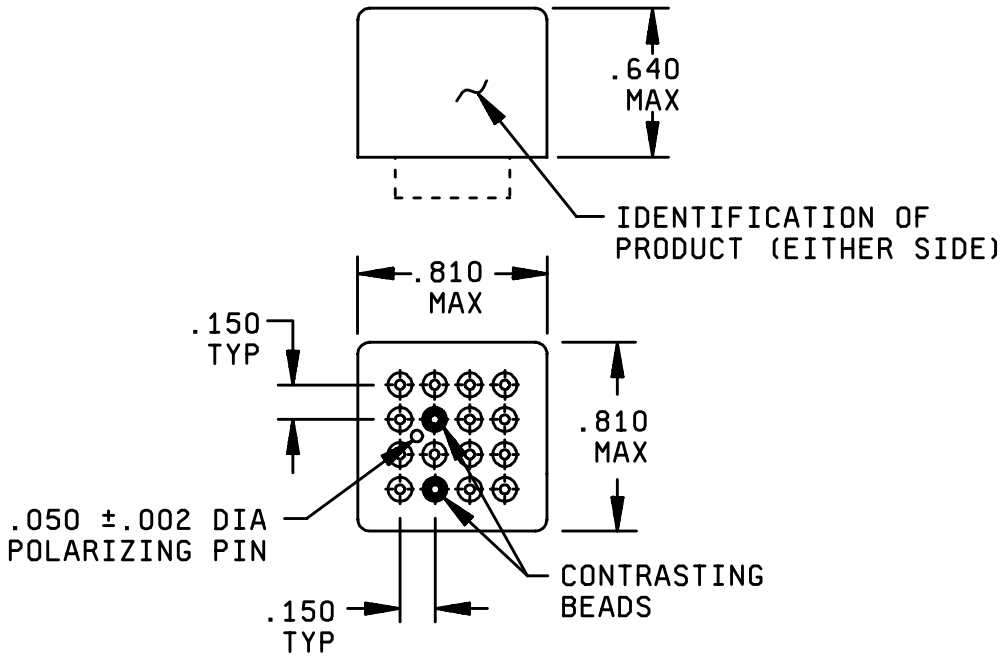
MIL-PRF-6106/39C(USAF)
30 July 1993
SUPERSEDING
MIL-R-6106/39B(USAF)
15 March 1985

PERFORMANCE SPECIFICATION SHEET

# RELAYS, ELECTROMAGNETIC, TYPE I, MAGNETIC LATCH, PERMANENT MAGNET DRIVE, LOW LEVEL TO 5 AMPERES, 4 PDT, HERMETICALLY SEALED

This specification is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification list in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation MIL-PRF-6106.

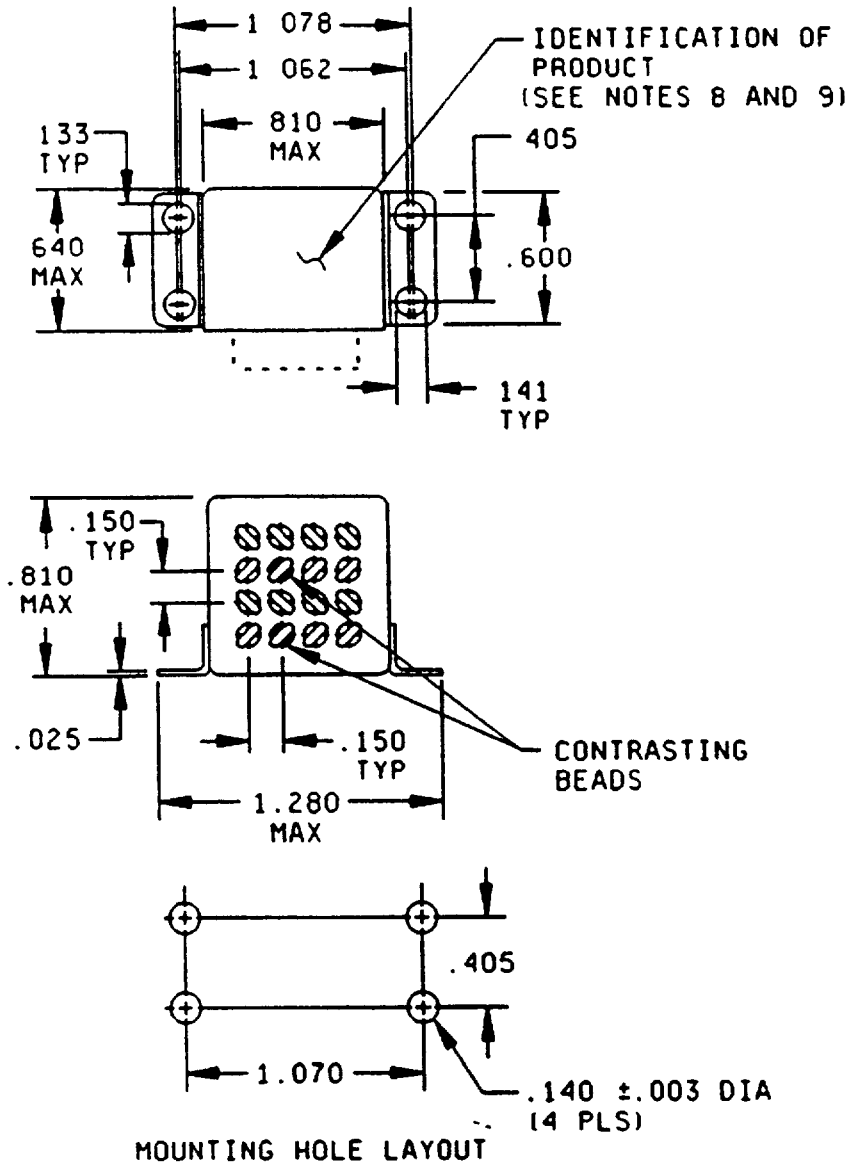



CONFIGURATION A

FIGURE 1. Dimensions and configurations.


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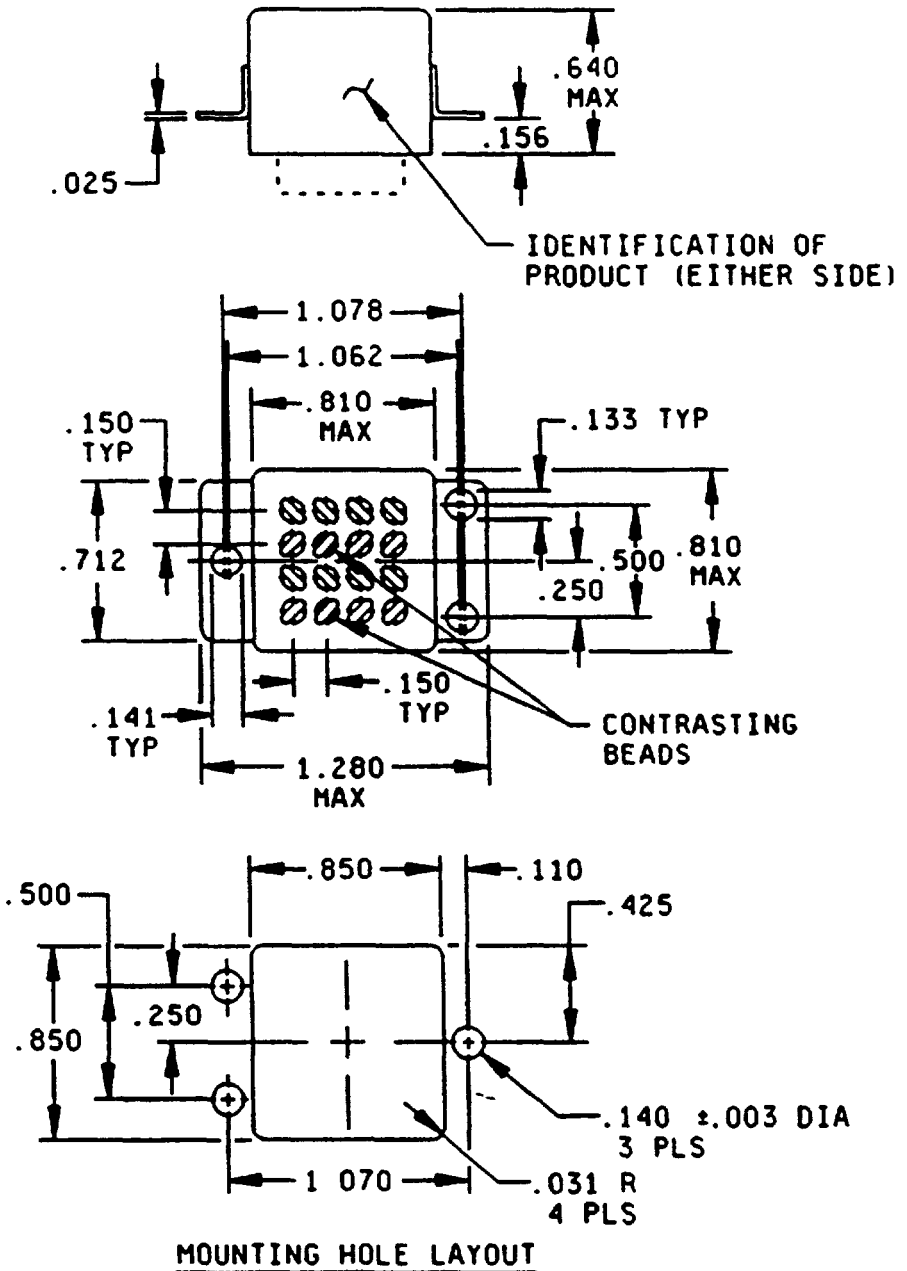

MOUNTING HOLE LAYOUT

CONFIGURATION B

FIGURE 1 Dimensions and configurations - Continued



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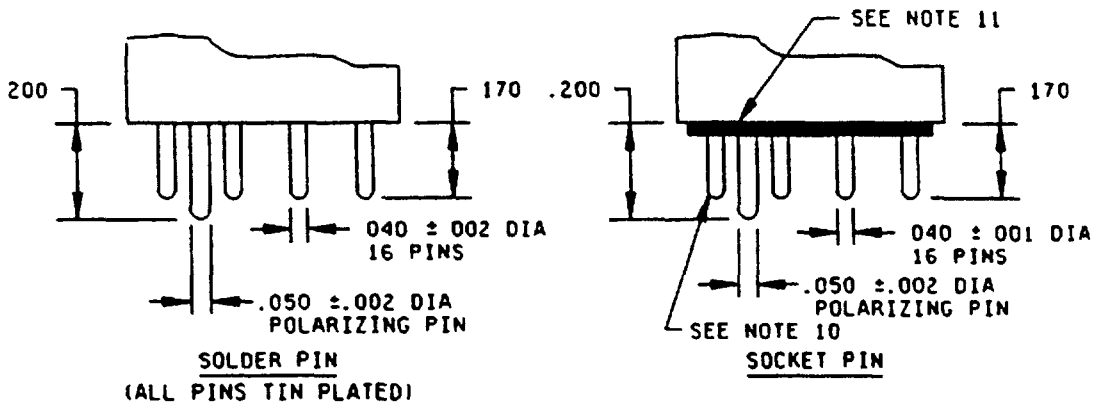

MOUNTING HOLE LAYOUT

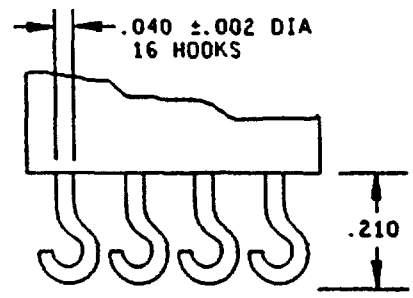
CONFIGURATION C

FIGURE 1 Dimensions and configurations - Continued

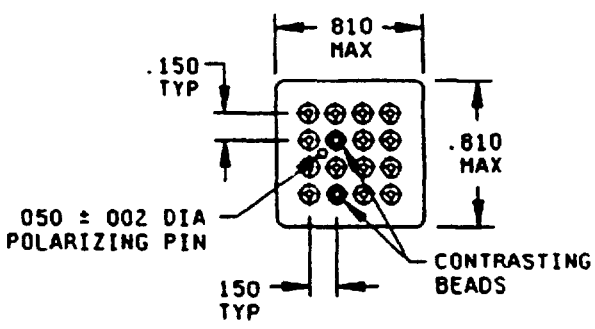


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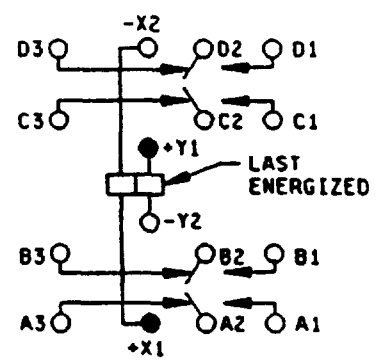



TERMINALS




TERMINAL VIEW
SOLDER PIN AND SOCKET PIN




CIRCUIT DIAGRAM

TO CLOSE NO. 1 CONTACTS (OPERATE)
ENERGIZE X1 AND X2

TO CLOSE NO. 3 CONTACTS (RESET)
ENERGIZE Y1 AND Y2

FIGURE 1 Dimensions and configurations - Continued



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| Inches | mm | Inches | mm | Inches | mm |
|---|---|---|---|---|---|
| 001 | 0 03 | 133 | 3 38 | 425 | 10 80 |
| 002 | 0 05 | 140 | 3 56 | .500 | 12 70 |
| 003 | 0 08 | 141 | 3 58 | 640 | 16 26 |
| 025 | 0.64 | .150 | 3.81 | 712 | 18.08 |
| 031 | 0 79 | 156 | 3.96 | 810 | 20 57 |
| 040 | 1 02 | 170 | 4.32 | .850 | 21 59 |
| 050 | 1.27 | .200 | 5.08 | 1 062 | 26.97 |
| 075 | 1 91 | 210 | 5 33 | 1.070 | 21 18 |
| 110 | 2 79 | 250 | 6.35 | 1.078 | 27 38 |
| | | | | 1 280 | 32 51 |

NOTES

1. Dimensions are in inches
2. Metric equivalents are given for general information only
3. Unless otherwise specified, tolerance is ± 010 inch (0.25 mm)
4. There shall be affixed to the relay a legible circuit diagram that identifies each terminal location specified
5. These relays are polarized monostable.
6. This relay shall not operate or be damaged by reverse polarity. Semiconductors shall not be used for this purpose
7. Permanent magnet drive consists of a permanent magnet with its flux path switched and combined with the electromagnetic flux
8. Applicable to configuration B only. The circuit diagram, manufacturer's PIN, and the military PIN shall be marked on the near side The remaining portion of the nameplate data shall be marked on the far side
9. Applicable to configuration B only. Relays shall be marked with the manufacturer's name or source code and date code In addition, ER relays shall be marked with the serial number Marking shall be with the bottom of the print adjacent to the near side
10. Socket pin terminals, except the polarizing pin, shall be gold-plated in accordance with MIL-G-45204, type II, class I. The nickel underplating shall be 50 to 150 microinches thick (minimum)
11. Silicone rubber gasket, AMS 3332, shore hardness 25 ±5, thickness .050 ±.005 inch (1 27 ±0 13 mm)

FIGURE 1 Dimensions and configurations - Continued



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REQUIREMENTS

CONTACT DATA

Life Unless otherwise specified, 100,000 cycles

Operational reliability Applicable, the alternate low level method shall be used

Load ratings

High level (relay case grounded)

Resistive 5 amperes at 28 V dc, 115 V ac, 400 Hz, 1 phase, 115/200 V ac, 400 Hz, 3 phase

Inductive 3 amperes at 28 V dc, 5 amperes at 115 V ac, 400 Hz, 1 phase; 115 V ac, 400 Hz, 3 phase (Life 20,000 cycles )

Motor 2 amperes at 28 V dc, 3 amperes at 115 V ac, 400 Hz, 1 phase, 115 V ac, 400 Hz, 3 phase

Lamp 1 ampere at 28 V dc, 1 ampere at 115 V ac, 400 Hz, 1 phase

Low level 10 to 50 μA at 10 to 50 mV dc or peak ac. 1/ 2/ 3/

Intermediate current: Applicable

Transfer load Not applicable

Mechanical life (reduced current) 1.25 amperes at 28 V dc, 115 V ac, 400 Hz, 1 phase, 115 V ac, 400 Hz, 3 phase (Life. 400,000 cycles )

Mixed loads Applicable 1/

Contact voltage drop or resistance 2/ 3/

High level·
Initial 0 150 volt maximum
After life· 0 175 volt maximum

Low level
Initial· 0 050 ohm maximum.
After life 0 150 ohm maximum

Contact bounce 0 5 millisecond maximum

---

1/ Relay shall be capable of switching low level while switching any of the other rated loads on adjacent contacts

2 For low level, the following shall apply
a During endurance and operational reliability testing, contact load shall be 10 to 50 microamperes, 10 to 50 millivolts open circuit voltage, 100 ohms maximum contact resistance
b Static contact resistance shall be performed at 50 mA maximum, 50 mV maximum

3/ For group A contact voltage drop test, high level testing shall be performed first, followed by low level testing The contacts shall not make or break the high level load



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Overload current 20 amperes dc
30 amperes ac

Rupture current 25 amperes dc.
40 amperes ac

Time current relay characteristics. Applicable (see table I)

TABLE I. Time current relay characteristics 1/ 2/

| | | |
|---|---|---|
| 1 | 10 A | 1 hour |
| 2 | 15 A | 20 seconds |
| 3 | 37 5 A | 2 seconds |
| 4 | 75 A | 53 second |

1/ CAUTION: Compare with time current characteristics of the associated circuit protective device

2/ All relays shall withstand overload and fault currents Relays must be able to sustain five applications (make and carry only) of power concurrently on adjacent poles at each of four different current levels in the sequence listed in table I Separate relays shall be tested at 28 V dc and 115/200 V ac, 400 Hz, 3 phase Cooling time between successive applications shall be 30 minutes. Tests shall be performed on both normally open and normally closed contacts There shall be no failures or evidence of welding or sticking and relays shall pass contact voltage drop at the conclusion

COIL DATA 4/: (See table II )

Operate time: 6 milliseconds maximum with rated coil voltage at 25°C. 7 milliseconds maximum with rated coil voltage over the temperature range

Duty rating: Continuous

ELECTRICAL DATA

Insulation resistance, initial: 100 megohms at 500 V dc

After life or environmental tests: 50 megohms at 500 V dc

| Dielectric withstanding voltage (sea level) | Initial V rms (60 Hz) | After life tests V rms (60 Hz) |
|---|---|---|
| Coil to case | 500 | 350 |
| Coil to coil | 500 | 350 |
| All other points | 1,000 | 750 |

| Dielectric withstanding voltage (altitude) 5/ | 80,000 feet V rms (60 Hz) | 300,000 feet V rms (60 Hz) |
|---|---|---|
| Coil to case | 250 | 500 |
| Coil to coil | 250 | 500 |
| All other points | 250 | 500 |

4/ CAUTION Due to possible interaction of relay magnetic fields, the following spacing requirements, as a minimum, shall be considered in dense packaging situations:
a Row to row assisting fields 125 inch
b Row to row opposing fields 1875 inch.
c Side to side alternating fields 0625 inch
d Side to side like fields 125 inch

5/ Dielectric may be improved by suitable insulation of terminals and wiring after installation



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TABLE II Dash numbers and characteristics 1/

| Military PIN M6106/39- | Terminal type | Mounting configuration | Coil data | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | | Rated voltage (V dc) 2/ | Resistance Ω ±10% at 25°C | Maximum | | Pickup voltage (max) (V dc) | | |
| | | | | | Voltage (V dc) 3/ | Current (A) at 25°C | Normal 4/ | High temp test | Continuing current test |
| 001 | Solder hook | C | 28 | 600 | 29 | 054 | 18 | 19 8 | 22.5 |
| 002 5/ | Socket pin | C | " | " | " | " | " | " | " |
| 003 | Solder pin | C | " | " | " | " | " | " | " |
| 004 | Solder hook | B | " | " | " | " | " | " | " |
| 005 | Solder pin | A | " | " | " | " | " | " | " |
| 006 | Solder pin | B | " | " | " | " | " | " | " |

1/ Each relay has high level and low level capabilities However, relays previously tested or used above 10 mA resistive at 6 V dc maximum or peak ac open circuits are not recommended for subsequent use in low level applications.

2/ CAUTION Use of any coil voltage less than rated coil voltage will compromise the operation of the relay

3/ When maximum ambient temperature does not exceed +85°C, the maximum coil voltage shall be 32 V dc

4/ Over the temperature range

5/ Use MIL-S-12883/40, PIN M12883/40-002, for mating socket

ENVIRONMENTAL CHARACTERISTICS

Temperature range -70°C to +125°C

Maximum altitude rating 300,000 feet

Acceleration Applicable (15 g's maximum)

Shock (specified pulse) MIL-STD-202, method 213, test condition C (100 g's), except configurations A and C peak g value shall be 200 g's. Contact chatter shall not exceed 10 microseconds maximum for closed contacts and 1 microsecond maximum for open contacts

Vibration (sinusoidal): Configurations A and C shall be 30 g's, 70 to 3,000 Hz Configuration B shall be 20 g's, 57 to 3,000 Hz Contact chatter shall not exceed 10 microseconds maximum for closed contacts and 1 microsecond maximum for open contacts.

Vibration (random) Applicable to qualification and group C inspection Test in accordance with MIL-STD-202, method 214, test condition IG (0 $4g^2/Hz$, 50 to 2,000 Hz), 15 minutes each plane, for horizontal flange mount test condition IE (0 $2g^2/Hz$) Contact chatter shall not exceed 10 microseconds maximum for closed contacts and 1 microsecond maximum closure for open contacts



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PHYSICAL DATA

Terminal strength

Solder hook terminals

Twist test  Not applicable

Pull force  5 ±0 5 pounds

Solder pin terminals

Bend test  Applicable

Pull force (0 040 diameter)  5 ±0.5 pounds

Pull force (0 050 diameter)  10 ±1 pounds (applicable to polarizing pin only)

Plug-in terminals

Bend test  Applicable

Pull force (0 040 diameter):  5 ±0.5 pounds

Pull force (0 050 diameter)  10 ±1 pounds (applicable to polarizing pin only)

Terminal solderability  Applicable to solder pin and solder hook terminals only.

Dimensions and configurations  See figure 1

Weight (maximum)·  0 064 pound (29 1 grams)

Seal  Hermetic, relays shall be sealed by welding ı e , laser, TIG (tungsten inert gas) or other suitable means as approved by the qualifying activity

Construction (internal and external)  All welded, except that coil magnet wire to coil lead wire shall be soldered

Identification of product  Applicable



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TABLE IV Qualification by similarity

| PIN M6106/39- | Basic qualification A (DC) | Dynamics Socket/pins | | | | Environmental A (DC) |
|---|---|---|---|---|---|---|
| | | 1/ A | B | C | D | |
| 001 | 4 | | Config C 2/ | | | 4 |
| 002 | 4 | Config C 2/ | | | | 4 2/ |
| 003 | 4 | | Config C 2/ | | | 4 |
| 004 | 4 | | | Config B 2/ | | 4 |
| 005 | 4 | | | | Config A 2/ | 4 |
| 006 | 4 | | | Config B 2/ | | 4 |

1/ Terminals Socket pin type terminals are considered worst case Solder pins and hooks may be qualified by similarity

2/ Socket pins (with gasket seals), are considered nonsimilar to solder pins and solder hooks. Each type must be tested

Revision letters are not used to denote changes due to the extensiveness of the changes

CONCLUDING MATERIAL

Custodian
Air Force - 85

Review activities
Air Force - 99
DLA - ES

Preparing activity
Air Force - 85

Agent
DLA - ES

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